

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD2323 FRONT-END WEB DEVELOPMENT /
UCCD2223 WEB SITE DESIGN AND DEVELOPMENT

MONDAY, 20 MAY 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A (Answer ALL Questions)

- Q1. (a) List **THREE (3)** HyperText Markup Language (HTML) block-level elements. (3 marks)
- (b) Describe the characteristics of block-level elements. (4 marks)
- (c) Write the HTML to display a web page with a video player that allows users to play the video file. The video file is saved in **TWO (2)** different file formats, "UTAR.m4v" and "UTAR.ogv". The video must also display the caption in English. The file containing the subtitle is "UTAR.vtt". The output is as shown in **Figure 1.1**.



Figure 1.1: UTAR video

- (d) Write the HTML character entity to display the following in a web page:
- (i) a "less than" sign (1 mark)
 - (ii) a registered trademark (1 mark)
 - (iii) an "ampersand" sign (1 mark)
 - (iv) a "copyright" sign (1 mark)
- (e) Write the HTML markup to create the web page that displays a nested list as shown in **Figure 1.2**.

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Q1. (e) (Continued)

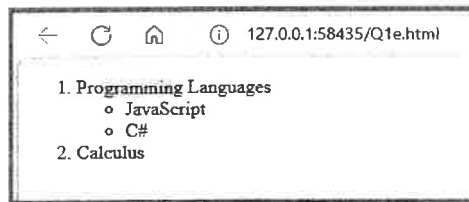


Figure 1.2: Nested list

(6 marks)

[Total : 25 marks]

Q2. (a) Explain with an example, the meaning of country-code top-level domain (ccTLD). (4 marks)

(b) Provide **ONE (1)** advantage and **ONE (1)** disadvantage of Cloud Hosting. (4 marks)

(c) What are the **THREE (3)** types of style sheets that can be used to apply Cascading Style Sheets (CSS) rules to HTML elements? Explain and provide an example for each of them. (9 marks)

(d) Write the CSS for the following:

(i) Set the font color to blue with 50%-transparent for all the div elements. (3 marks)

(ii) Draw black border for the p elements that are descendants of div elements. (3 marks)

(e) Explain the meaning of a class selector. (2 marks)

[Total : 25 marks]

Q3. (a) JavaScript code may appear in the <head> section or in the <body> section.

(i) Explain how to indicate to the browser that the text that follows is part of a script. (2 marks)

(ii) Explain with examples how comments can be added to the JavaScript code to provide explanations or context. (6 marks)

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Q3. (Continued)

- (b) Write a JavaScript code to display the output as shown in **Figure 3.1**. (6 marks)

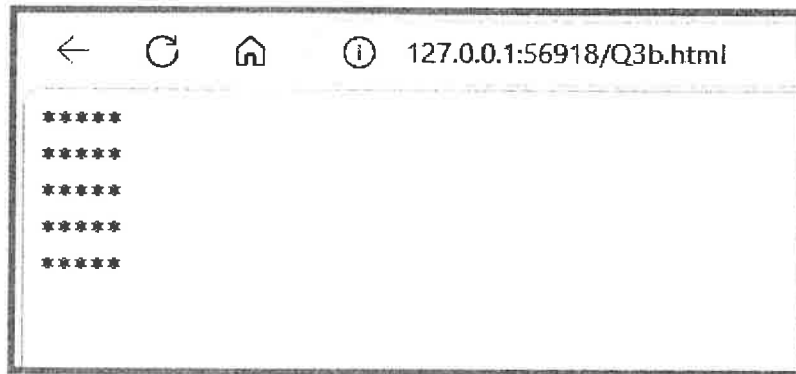


Figure 3.1: Sample input

- (c) (i) Write the code for a JavaScript function that takes two arguments x and y that are floating point numbers and returns the product of the two arguments. [Note: Name the function as product.] (4 marks)
- (ii) Write the JavaScript code to call the product function created in (c)(i) with 7.2 and 3.5 as arguments. The result will be displayed in the web browser in 2 decimal places. (4 marks)
- (d) Provide **THREE (3)** reasons for writing functions. (3 marks)

[Total : 25 marks]

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Section B (Answer only ONE (1) question)

Q4. (a) The HTML markup for a web page is as shown in **Figure 4.1**.

```
<form id="splitnameform">
  <p>Enter your name:</p>
  <p>
    <input id="name" type="text">
    <input type="submit" value="Split Name" </p>
    <div id="results"></div>
  </form>
```

Figure 4.1: HTML markup

Write the JavaScript code that breaks the name into words and then calculate the average number of characters per word. The results will be displayed in `<div id="results">`. (10 marks)

(b) With an example, explain how the expiration date of a cookie can be configured. (6 marks)

(c) Differentiate between session storage and local storage. (6 marks)

(d) Define event handler. (3 marks)

[Total : 25 marks]

Q5. (a) Provide **THREE (3)** ways to create a custom object. (6 marks)

(b) The variable “grades” is an array that is used to store the mathGrade, biologyGrade and chemistryGrade objects. All the three grade objects have the properties called “Subject” and “Marks”. Write the JavaScript code to store the information in **Table 5.1** into the Grades array.

Subject	Marks
Maths	80
Biology	90
Chemistry	70

Table 5.1: Grades obtained

(5 marks)

(c) Write the JavaScript code to store the grades array in Session storage. (3 marks)

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Q5. (Continued)

- (d) Write the JavaScript code to retrieve the grades from Session storage. (3 marks)
- (e) The HTML markup is as shown in **Figure 5.1**.

```
<form>
  <p>First Date: <input id="firstDate" name="firstDate" type="date"></p>
  <p>Second Date: <input id="secondDate" name="secondDate" type="date"></p>
  <p>
    <input type="submit" value="Compare dates"> <br>
    <input type="reset" value="Clear Dates">
  </p>
</form>
<div id="results"></div>
```

Figure 5.1: HTML markup

Write the JavaScript code to compare the First Date and the Second Date and display either one of the following messages in `<div id="results">`:

- "First Date is greater than Second Date"
- "Second Date is greater than First Date"
- "Both dates are equal."

(8 marks)

[Total : 25 marks]